

Bitácora de comandos: revisión piloto de BAMs del proyecto Macandog et al.

Objetivo: Observar lo que se encuentra en los BAMs del *Proyecto ENA PRJEB61942 / ERP147020 · BAM AMPLICON/16S y BAM WGS/shotgun*

Resumen de conclusión

BAM	Tipo usado como piloto	Lecturas totales	Pares (Read1/Read2)	Longitud promedio	Mapped	Conclusión
ERR13828068.bam	AMPLICON / 16S	224,904	112,452	251 bp	0.00%	No alineado
ERR13877920.bam	WGS / shotgun	63,054,758	31,527,379	148 bp	0.00%	No alineado

Son BAMs de lecturas no alineadas.

1. Directorio de trabajo y descarga de BAMs

Directorio de trabajo:

```
cd /mnt/data/sur/users/abadia/macandog_bam_pilot
pwd
```

2. Job de descarga con SLURM

Enlaces usados para descargar los dos BAM de prueba:

```
#!/bin/bash
#SBATCH --job-name=macandog_download
#SBATCH --partition=interactive
#SBATCH --time=08:00:00
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G
#SBATCH --output=download_%j.out
#SBATCH --error=download_%j.err

set -euo pipefail

cd /mnt/data/sur/users/abadia/macandog_bam_pilot

echo "Iniciando WGS: $(date)"

wget -c --tries=0 --timeout=60 -o ERR13877920.wget.log "https://ftp.sra.ebi.ac.uk/vol1/err/ERR138/020/ERR13877920/ERR13877920.bam"

echo "WGS terminado: $(date)"
echo "Iniciando 16S: $(date)"
```

```
wget -c --tries=0 --timeout=60 -o ERR13828068.wget.log "https://ftp.sra.ebi.ac.uk/vol1/err/ERR138/068/ERR13828068/ERR13828068.bam"

echo "16S terminado: $(date)"
echo "Descargas terminadas."

ls -lh ERR13877920.bam ERR13828068.bam
```

Envío del job:

```
sbatch download_bams.sbatch
```

Estado observado del job:

```
237696 interacti macandog abadia R 0:17 1 node12
```

3. Verificación de archivos descargados

Al terminar la descarga, el directorio contenía:

```
[abadia@login01 macandog_bam_pilot]$ ls
ERR13828068.bam      ERR13877920.wget.log  download_bams.sbatch
ERR13828068.wget.log download_237696.err
ERR13877920.bam      download_237696.out
```

4. Carga de samtools

```
[abadia@node12 macandog_bam_pilot]$ module load samtools/1.22.1
```

5. quickcheck inicial

Sacar información breve:

```
[abadia@node12 macandog_bam_pilot]$ samtools quickcheck -v ERR13828068.bam
ERR13828068.bam had no targets in header.
ERR13828068.bam
[abadia@node12 macandog_bam_pilot]$ samtools quickcheck -v ERR13877920.bam
ERR13877920.bam had no targets in header.
ERR13877920.bam
```

Samtools avisó que los BAM no tienen targets en el header.

Después se corrió quickcheck aceptando BAMs no alineados con -u:

```
[abadia@node12 macandog_bam_pilot]$ samtools quickcheck -u -v ERR13828068.bam
samtools quickcheck -u -v ERR13877920.bam
```

Salida: no imprimió errores.

6. Header: búsqueda de referencias @SQ

```
[abadia@node12 macandog_bam_pilot]$ mkdir -p reports

samtools view -H ERR13828068.bam > reports/ERR13828068.header.sam
samtools view -H ERR13877920.bam > reports/ERR13877920.header.sam

grep -c '^@SQ' reports/ERR13828068.header.sam
grep -c '^@SQ' reports/ERR13877920.header.sam
0
0
```

Ambos BAMs tienen 0 referencias @SQ. Por tanto, no están alineados contra ninguna referencia.

7. flagstat

Comandos usados:

```
[abadia@node12 macandog_bam_pilot]$ samtools flagstat -@ 4 ERR13828068.bam > reports/ERR13828068.flagstat.txt
samtools flagstat -@ 4 ERR13877920.bam > reports/ERR13877920.flagstat.txt

cat reports/ERR13828068.flagstat.txt
cat reports/ERR13877920.flagstat.txt
```

Salida de ERR13828068.bam:

```
224904 + 0 in total (QC-passed reads + QC-failed reads)
224904 + 0 primary
0 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
0 + 0 mapped (0.00% : N/A)
0 + 0 primary mapped (0.00% : N/A)
224904 + 0 paired in sequencing
112452 + 0 read1
112452 + 0 read2
0 + 0 properly paired (0.00% : N/A)
0 + 0 with itself and mate mapped
0 + 0 singletons (0.00% : N/A)
0 + 0 with mate mapped to a different chr
0 + 0 with mate mapped to a different chr (mapQ>=5)
```

Salida de ERR13877920.bam:

```
63054758 + 0 in total (QC-passed reads + QC-failed reads)
63054758 + 0 primary
0 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
```

```

0 + 0 mapped (0.00% : N/A)
0 + 0 primary mapped (0.00% : N/A)
63054758 + 0 paired in sequencing
31527379 + 0 read1
31527379 + 0 read2
0 + 0 properly paired (0.00% : N/A)
0 + 0 with itself and mate mapped
0 + 0 singletons (0.00% : N/A)
0 + 0 with mate mapped to a different chr
0 + 0 with mate mapped to a different chr (mapQ>=5)

```

Interpretación: ambos BAMs tienen lecturas paired-end, pero 0 lecturas mapeadas. Esto confirma que no son alineamientos.

8. stats

Comandos usados:

```

[abadia@node12 macandog_bam_pilot]$ samtools stats -@ 4 ERR13828068.bam > reports/ERR13828068.stats.txt
samtools stats -@ 4 ERR13877920.bam > reports/ERR13877920.stats.txt

grep '^SN' reports/ERR13828068.stats.txt | head -n 40
grep '^SN' reports/ERR13877920.stats.txt | head -n 40

```

Salida de ERR13828068.bam:

```

SN      raw total sequences: 224904 # excluding supplementary and secondary reads
SN      filtered sequences: 0
SN      sequences: 224904
SN      is sorted: 1 # sorted by coordinate
SN      1st fragments: 112452
SN      last fragments: 112452
SN      reads mapped: 0
SN      reads mapped and paired: 0 # paired-end technology bit set + both mates mapped
SN      reads unmapped: 224904
SN      reads properly paired: 0 # proper-pair bit set
SN      reads paired: 224904 # paired-end technology bit set
SN      reads duplicated: 0 # PCR or optical duplicate bit set
SN      reads MQ0: 0 # mapped and MQ=0
SN      reads QC failed: 0
SN      non-primary alignments: 0
SN      supplementary alignments: 0
SN      total length: 56450904 # ignores clipping
SN      total first fragment length: 28225452 # ignores clipping
SN      total last fragment length: 28225452 # ignores clipping
SN      bases mapped: 0 # ignores clipping
SN      bases mapped (cigar): 0 # more accurate
SN      bases trimmed: 0
SN      bases duplicated: 0
SN      mismatches: 0 # from NM fields
SN      error rate: 0.000000e+00 # mismatches / bases mapped (cigar)
SN      average length: 251

```

SN	average first fragment length:	251
SN	average last fragment length:	251
SN	maximum length:	251
SN	maximum first fragment length:	251
SN	maximum last fragment length:	251
SN	average quality:	35.1
SN	insert size average:	0.0
SN	insert size standard deviation:	0.0
SN	inward oriented pairs:	0
SN	outward oriented pairs:	0
SN	pairs with other orientation:	0
SN	pairs on different chromosomes:	0
SN	percentage of properly paired reads (%)	0.0

Salida de ERR13877920.bam:

SN	raw total sequences:	63054758	# excluding supplementary and secondary reads
SN	filtered sequences:	0	
SN	sequences:	63054758	
SN	is sorted:	1	# sorted by coordinate
SN	1st fragments:	31527379	
SN	last fragments:	31527379	
SN	reads mapped:	0	
SN	reads mapped and paired:	0	# paired-end technology bit set + both mates mapped
SN	reads unmapped:	63054758	
SN	reads properly paired:	0	# proper-pair bit set
SN	reads paired:	63054758	# paired-end technology bit set
SN	reads duplicated:	0	# PCR or optical duplicate bit set
SN	reads MQ0:	0	# mapped and MQ=0
SN	reads QC failed:	0	
SN	non-primary alignments:	0	
SN	supplementary alignments:	0	
SN	total length:	9309387822	# ignores clipping
SN	total first fragment length:	4654605795	# ignores clipping
SN	total last fragment length:	4654782027	# ignores clipping
SN	bases mapped:	0	# ignores clipping
SN	bases mapped (cigar):	0	# more accurate
SN	bases trimmed:	0	
SN	bases duplicated:	0	
SN	mismatches:	0	# from NM fields
SN	error rate:	0.000000e+00	# mismatches / bases mapped (cigar)
SN	average length:	148	
SN	average first fragment length:	148	
SN	average last fragment length:	148	
SN	maximum length:	151	
SN	maximum first fragment length:	151	
SN	maximum last fragment length:	151	
SN	average quality:	35.3	
SN	insert size average:	0.0	
SN	insert size standard deviation:	0.0	
SN	inward oriented pairs:	0	
SN	outward oriented pairs:	0	

SN	pairs with other orientation:	0
SN	pairs on different chromosomes:	0
SN	percentage of properly paired reads (%) :	0.0